

29/7/26

Roshan .R
192524085
CSA0313

CO-2 Assessment Pool - 1

Analytical Problem Solving

1. a) Initial List :

L1 = Empty

Current : |

L1. append (10)

10 $\rightarrow$ |

L1. append (20)

10 $\rightarrow$ 20 $\rightarrow$ |

L1. append (15)

10 $\rightarrow$ 20 $\rightarrow$ 15 $\rightarrow$ |

Final List :

 $\langle | 10, 20, 15 \rangle$

(Current position is at 10)

b) Initial Configuration :

L2. append (10);

List state $\langle | 10 \rangle$

L2. append (20);

List state $\langle | 10, 20 \rangle$

L2. append (15);

List state $\langle |10, 20, 15 \rangle$

L2. moveToStart();

List state $\langle |10, 20, 15 \rangle$

(Current position explicitly set to index 0)

L2. insert (39);

→ Inserts 39 at the current position. 10 shifts right, & the current position remains at newly inserted element.

L2. next();

→ cursor moves forward one position to 10.

List state: $\langle 39, |10, 20, 15 \rangle$

L2. insert (12);

Inserts 12 at the current position

List state

$\langle 39, |12, 10, 20, 15 \rangle$

Final Configuration

$\langle 39, |12, 10, 20, 15 \rangle$

(Current position is at 12)

2. Binary Search Algorithm :-

i) Maintain two pointers low (starting index, 0) & high (ending index, $N-1$).

ii) while $low \leq high$:

→ Calculate mid index

$$mid = \frac{low + high}{2}$$

→ Match if $Array[mid] == key$,
return mid.

→ Key is smaller, search the left half by updating $high = mid - 1$.

→ Key is larger, search the right half by updating $low = mid + 1$.

iii) If $low > high$, the target element is not present in array (return -1).

Given dataset

Index	0	1	2	3	4	5	6	7	8	9
value	66	111	212	323	432	543	643	649	777	888

i) Key = 111

low = 0, high = 9

$$mid = \frac{9}{2} = 4 = Array[4] = 432$$

$111 < 432 \Rightarrow$ Search left half :
 $high = 4 - 1 = 3$

* low = 0, high = 3

mid = $\frac{3}{2} = 1 \Rightarrow \text{Array}[1] = 111$

$111 = 111 \Rightarrow \text{found}$

$\therefore$ key 111 found at index 1 in 2 iterations

ii) key = 888

* low = 0, high = 9

mid = $\frac{9}{2} = 4 \Rightarrow \text{Array}[4] = 432$

$888 > 432 \Rightarrow \text{Search right half}$

low = $4 + 1 = 5$

* low = 5, high = 9

mid = $\frac{14}{2} = 7 \Rightarrow \text{Array}[7] = 649$

$888 > 649 \Rightarrow \text{Search right half}$

low = $7 + 1 = 8$

* low = 8, high = 9

mid = $\frac{17}{2} = 8 \Rightarrow \text{Array}[8] = 777$

$888 > 777$

low = $8 + 1 = 9$

* low = 9, high = 9

mid = $\frac{18}{2} = 9 \Rightarrow \text{Array}[9] = 888$

$888 = 888 \Rightarrow \text{found}$

$\therefore$ key 888 is found at index 9 (4 iterations)

iii) key = 999

* low = 0, high = 9

mid = 4 $\Rightarrow$ 999 > 432

low = 5

* low = 5, high = 9

mid = 7 $\Rightarrow$ 999 > 647

low = 8

* low = 8, high = 9

mid = 8 $\Rightarrow$ 999 > 777

low = 9

* low = 9, high = 9

mid = 9 $\Rightarrow$ 999 > 888

low = 10

* Termination:

low = 10, high = 9

Since low > high, the loop terminates

$\therefore$ key 999 not found in array (return -1).